

Université Sorbonne Paris Nord
Ecole d'Ingénieurs Sup Galilée

TP1

Réalisé par:

BRAHIMI Kaoutar

Exercice n° 1:

1-

```
create table section
(course_id      varchar(8),
 sec_id        varchar(8),
 semester      varchar(6)
 check (semester in ('Fall', 'Winter', 'Spring', 'Summer')),
 year          numeric(4,0),
 building      varchar(15),
 room_number   varchar(7),
 time_slot_id  varchar(4),
 primary key (course_id, sec_id, semester, year),
 foreign key (course_id) references course,
 foreign key (building, room_number) references classroom
);
```

4- Insérer un nouveau cours dont l'identifiant est BIO-101, intitulé Intro. to Biology, assurée par le département Biology et son crédit est de 4.

```
insert into course values ('BIO-101', 'Intro. to Biology', 'Biology', '4');
```

Exercice n° 2:

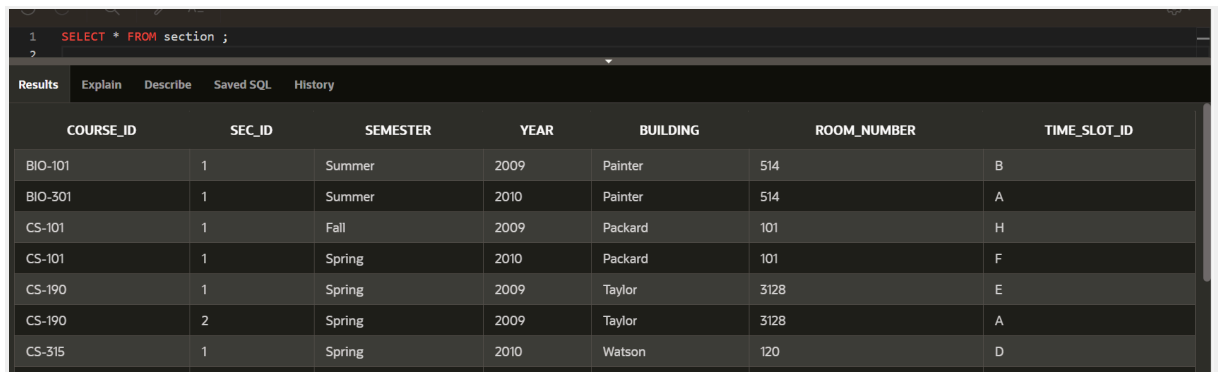
1. Afficher la structure de la relation section et son contenu.

```
DESC section ;
```

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SECTION	COURSE_ID	VARCHAR2	8	-	-	1	-	-	-
	SEC_ID	VARCHAR2	8	-	-	2	-	-	-
	SEMESTER	VARCHAR2	6	-	-	3	-	-	-
	YEAR	NUMBER	-	4	0	4	-	-	-
	BUILDING	VARCHAR2	15	-	-	-	✓	-	-
	ROOM_NUMBER	VARCHAR2	7	-	-	-	✓	-	-
	TIME_SLOT_ID	VARCHAR2	4	-	-	-	✓	-	-

```
SELECT *
```

FROM section ;



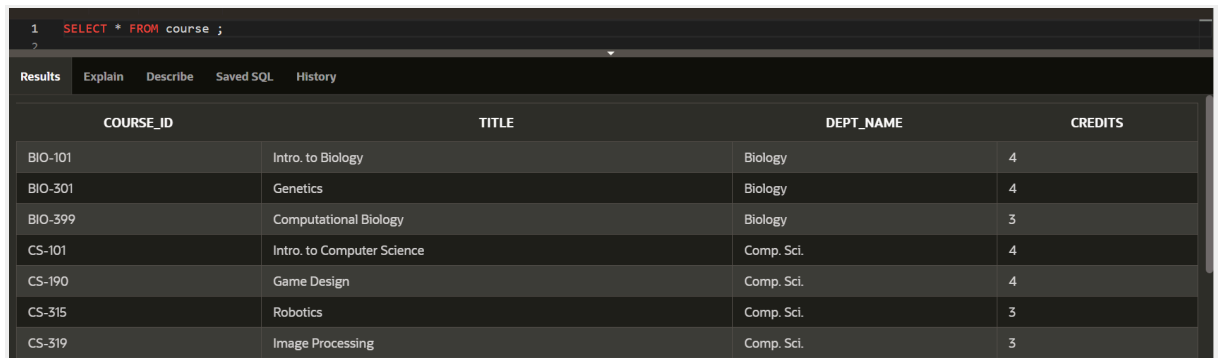
```
1 SELECT * FROM section ;
2
```

COURSE_ID	SEC_ID	SEMESTER	YEAR	BUILDING	ROOM_NUMBER	TIME_SLOT_ID
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D

2. Afficher tous les renseignements sur les cours que l'on peut programmer (relation course).

SELECT *

FROM course ;



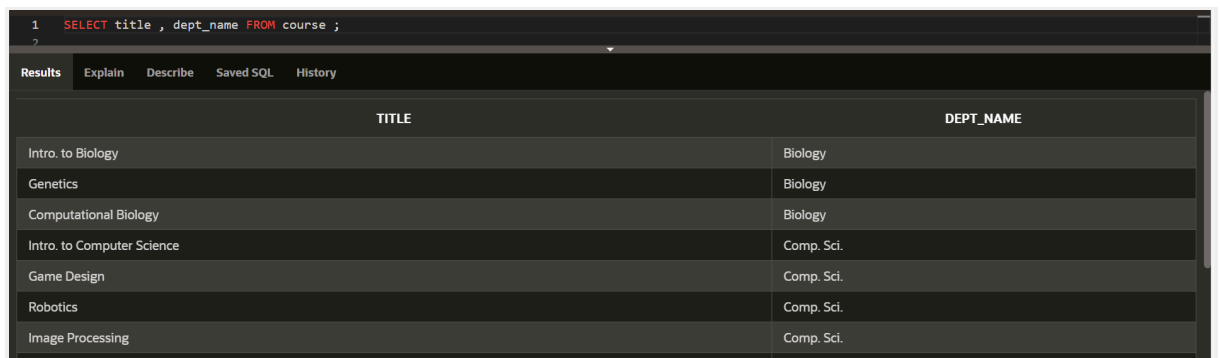
```
1 SELECT * FROM course ;
2
```

COURSE_ID	TITLE	DEPT_NAME	CREDITS
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3
CS-319	Image Processing	Comp. Sci.	3

3. Afficher les titres des cours et les départements qui les proposent.

SELECT title , dept_name

FROM course ;



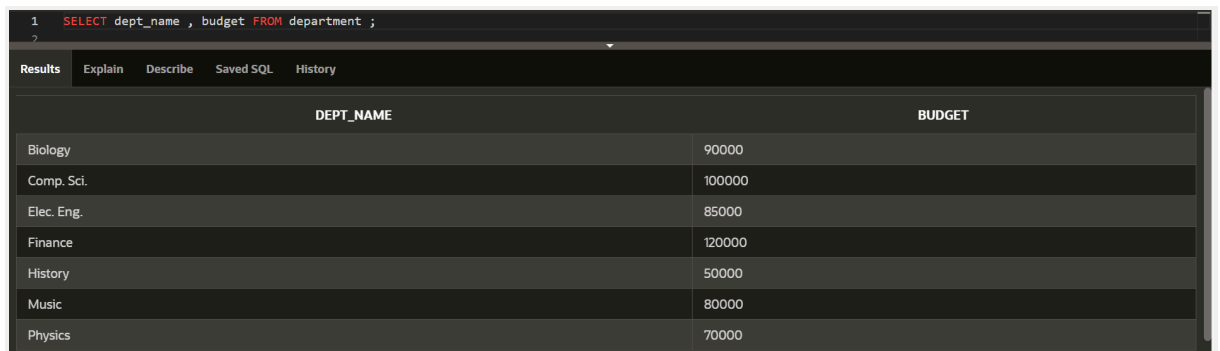
```
1 SELECT title , dept_name FROM course ;
2
```

TITLE	DEPT_NAME
Intro. to Biology	Biology
Genetics	Biology
Computational Biology	Biology
Intro. to Computer Science	Comp. Sci.
Game Design	Comp. Sci.
Robotics	Comp. Sci.
Image Processing	Comp. Sci.

4. Afficher les noms des départements ainsi que leur budget.

```
SELECT dept_name , budget
```

```
FROM department ;
```



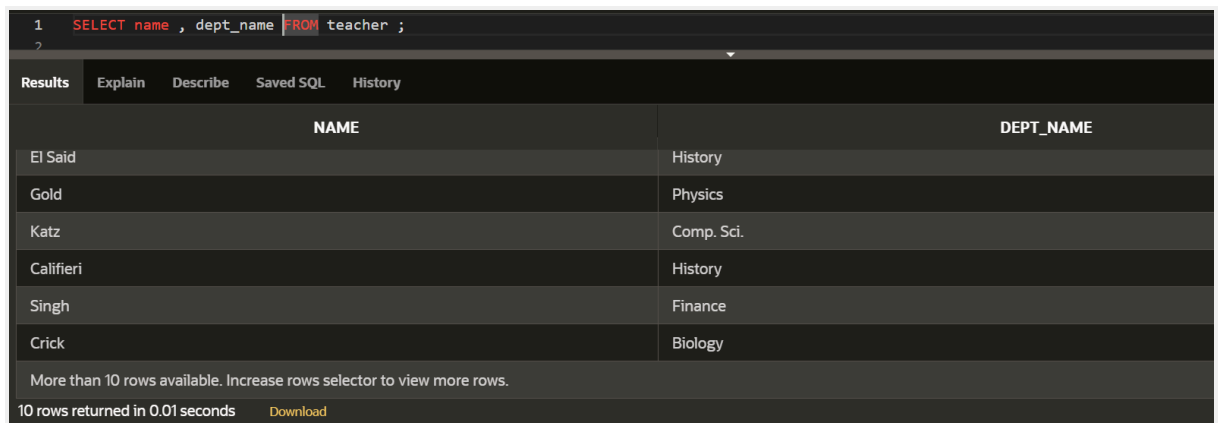
The screenshot shows a SQL query execution interface. The query is: `1 SELECT dept_name , budget FROM department ;`. The results are displayed in a table with two columns: **DEPT_NAME** and **BUDGET**. The data is as follows:

DEPT_NAME	BUDGET
Biology	90000
Comp. Sci.	100000
Elec. Eng.	85000
Finance	120000
History	50000
Music	80000
Physics	70000

5. Afficher tous les noms des enseignants et leur département.

```
SELECT name , dept_name
```

```
FROM teacher ;
```



The screenshot shows a SQL query execution interface. The query is: `1 SELECT name , dept_name FROM teacher ;`. The results are displayed in a table with two columns: **NAME** and **DEPT_NAME**. The data is as follows:

NAME	DEPT_NAME
El Said	History
Gold	Physics
Katz	Comp. Sci.
Califieri	History
Singh	Finance
Crick	Biology

More than 10 rows available. Increase rows selector to view more rows.
10 rows returned in 0.01 seconds [Download](#)

6. Afficher tous les noms des enseignants ayant un salaire supérieur strictement `a 65.000 \$.

```
SELECT name
```

```
FROM teacher
```

```
WHERE salary > 65000;
```

```
1 SELECT name FROM teacher WHERE salary > 65000;
```

NAME
Einstein
Gold
Katz
Singh
Crick
Brandt
Kim

8 rows returned in 0.01 seconds [Download](#)

7. Afficher les noms des enseignants ayant un salaire compris entre 55.000 \$ et 85.000 \$.

```
SELECT name
```

```
FROM teacher
```

```
WHERE salary between 55000 and 85000;
```

```
1 SELECT name FROM teacher WHERE salary between 55000 and 85000;
```

NAME
Srinivasan
El Said
Katz
Califieri
Singh
Crick
Kim

7 rows returned in 0.01 seconds [Download](#)

8. Afficher les noms des départements, en utilisant la relation teacher et éliminer les doublons.

```
SELECT DISTINCT dept_name
```

```
FROM teacher ;
```

```
1 SELECT DISTINCT dept_name FROM teacher ;
```

DEPT_NAME
Comp. Sci.
Biology
History
Finance
Elec. Eng.
Music
Physics

7 rows returned in 0.01 seconds [Download](#)

9. Afficher tous les noms des enseignants du département informatique ayant un salaire supérieur strictement à 65.000 \$.

```
select name
```

```
from teacher
```

```
where salary > 65000 and dept_name='Comp. Sci.' ;
```

```
1 select name from teacher where salary > 65000 and dept_name='Comp. Sci.' ;
```

NAME
Katz
Brandt

2 rows returned in 0.01 seconds [Download](#)

10. Afficher tous les renseignements sur les cours proposés au printemps 2010 (relation section).

```
select *
```

```
from section
```

```
where semester='Spring';
```

1 `select * from section where semester='Spring';`

Results

ExplainDescribeSaved SQLHistory

COURSE_ID	SEC_ID	SEMESTER	YEAR	BUILDING	ROOM_NUMBER	TIME_SLOT_ID
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D

10 rows returned in 0.01 secondsDownload

11. Afficher tous les titres des cours dispensés par le département informatique qui ont plus de trois crédits.

```
select title
```

```
from course
```

```
where dept_name ='Comp. Sci.' and credits > 3;
```

```
1 select title from course where dept_name = 'Comp. Sci.' and credits > 3;
```

Results	Explain	Describe	Saved SQL	History
TITLE				
Intro. to Computer Science				
Game Design				

2 rows returned in 0.00 seconds [Download](#)

12. Afficher tous les noms des enseignants ainsi que le nom de leur département et les noms des qui les hébergent.

```
SELECT teacher . name , teacher . dept_name , department . building
```

```
FROM teacher , department
```

```
WHERE teacher . dept_name = department . dept_name ;
```

```

1 SELECT teacher .name , teacher . dept_name , department . building
2 FROM teacher , department WHERE teacher . dept_name = department . dept_name ;

```

NAME	DEPT_NAME	BUILDING
Gold	Physics	Watson
Katz	Comp. Sci.	Taylor
Califieri	History	Painter
Singh	Finance	Painter
Crick	Biology	Watson

More than 10 rows available. Increase rows selector to view more rows.
10 rows returned in 0.01 seconds [Download](#)

13. Afficher tous les étudiants ayant suivi au moins un cours en informatique.

```

select distinct student.name
from student , takes, course
where student.id=takes.id and takes.course_id=course.course_id and course.dept_name =
'Comp. Sci.';

```

```

1 t distinct student.name from student , takes, course where student.id=takes.id and takes.course_id=course.course_id and course.dept_name = 'Comp. Sci.';

```

NAME
Brown
Zhang
Levy
Bourikas
Shankar
Williams

6 rows returned in 0.05 seconds [Download](#)

14. Afficher les noms des étudiants ayant suivi un cours dispensé par un enseignant nommé Eistein

```

SELECT DISTINCT student.name
FROM student, teacher, takes, teaches
WHERE student.id = takes.id
AND takes.course_id = teaches.course_id
AND takes.sec_id = teaches.sec_id
AND takes.semester = teaches.semester
AND takes.year = teaches.year

```



```
AND teaches.id = teacher.id
```

```
AND teacher.name = 'Einstein';
```

Results	Explain	Describe	Saved SQL	History
NAME				
Peltier				
1 rows returned in 0.04 seconds Download				

15. Afficher tous les identifiants des cours et les enseignants qui les ont assurés.

```
select course_id
```

```
from teacher , teaches
```

```
where teacher.id=teaches.id ;
```

1	select course_id from teacher , teaches where teacher.id=teaches.id ;			
Results	Explain	Describe	Saved SQL	History
COURSE_ID				
PHY-101				
HIS-351				
CS-101				
CS-319				
BIO-101				
More than 10 rows available. Increase rows selector to view more rows.				
10 rows returned in 0.01 seconds Download				

16. Afficher le nombre d'inscrits pour chaque enseignement proposé au printemps 2010.

```
select count(takes.id)
```

```
from takes
```

```
where takes.year=2010 and takes.semester='Spring';
```

```
1 select count(takes.id) from takes where takes.year=2010 and takes.semester='Spring';
```

Results	Explain	Describe	Saved SQL	History
COUNT(TAKES.ID)				
8				
1 rows returned in 0.03 seconds Download				

17. Afficher les noms des départements et les salaires maximums de leurs enseignants.

```
SELECT dept_name , max ( salary ) FROM teacher
GROUP BY dept_name ;
```

```
1 SELECT dept_name , max ( salary ) FROM teacher
2 GROUP BY dept_name ;
```

Results	Explain	Describe	Saved SQL	History
DEPT_NAME		MAX(SALARY)		
Biology	72000			
History	62000			
Finance	90000			
Elec. Eng.	80000			
Music	40000			
Physics	95000			
7 rows returned in 0.00 seconds Download				

18. Afficher le nombre d'inscrits pour chaque enseignement proposé.

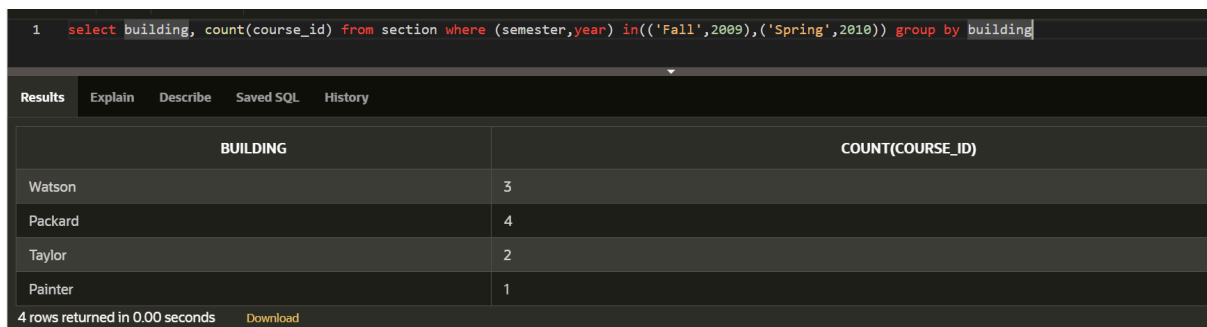
```
select course_id , count(id)
from takes group by course_id;
```

```
1 select course_id , count(id) from takes group by course_id;
```

Results	Explain	Describe	Saved SQL	History
COURSE_ID		COUNT(ID)		
CS-319	2			
CS-347	2			
PHY-101	1			
EE-181	1			
CS-315	2			
More than 10 rows available. Increase rows selector to view more rows.				
10 rows returned in 0.01 seconds Download				

19. On affiche le nombre total de cours qui ont eu lieu dans chaque bâtiment, pendant l'automne 2009 et le printemps 2010.

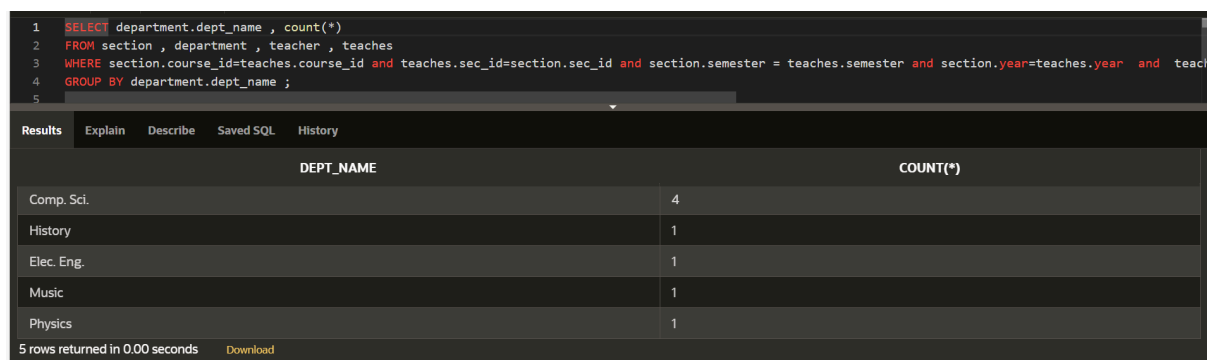
```
select building, count(course_id)
from section
where (semester,year) in(('Fall',2009),('Spring',2010)) group by building
```



BUILDING	COUNT(COURSE_ID)
Watson	3
Packard	4
Taylor	2
Painter	1

20. On affiche le nombre total de cours dispensés par chaque département et qui ont eu dans le même bâtiment qui l'abrite.

```
SELECT department.dept_name , count(*)
FROM section , department , teacher , teaches
WHERE section.course_id=teaches.course_id and teaches.sec_id=section.sec_id and
section.semester = teaches.semester and section.year=teaches.year and teaches.id =
teacher.id and teacher.dept_name = department.dept_name and
department.building=section.building
GROUP BY department.dept_name ;
```



DEPT_NAME	COUNT(*)
Comp. Sci.	4
History	1
Elec. Eng.	1
Music	1
Physics	1

21. On affiche les titres des cours proposés et qui ont eu lieu et les enseignants qui les ont assurés.

```
SELECT course.title , teacher.name
FROM section , department , teacher , teaches , course
```

```
WHERE section.course_id=teaches.course_id and teaches.sec_id=section.sec_id and
section.semester = teaches.semester and section.year=teaches.year and teaches.id =
teacher.id and
section.course_id = course.course_id ;
```

```
1 SELECT course.title , teacher.name
2 FROM section , department , teacher , teaches , course
3 WHERE section.course_id=teaches.course_id and teaches.sec_id=section.sec_id and section.semester = teaches.semester and section.year=teaches.year and teaches.id = teacher.id and section.course_id = course.course_id ;
```

TITLE	NAME
Intro. to Biology	Crick
Genetics	Crick
Genetics	Crick
Genetics	Crick

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.04 seconds [Download](#)

22. On affiche le nombre total de cours qui ont eu lieu pour chacune des période Summer, Fall et Spring.

```
SELECT section.semester, count(*)
FROM section
GROUP BY section.semester;
```

```
1 SELECT section.semester, count(*) FROM section GROUP BY section.semester;
```

SEMESTER	COUNT(*)
Fall	3
Summer	2
Spring	10

3 rows returned in 0.00 seconds [Download](#)

23. On affiche pour chaque étudiant le nombre total de crédits qu'il a obtenu, en suivant des cours qui n'ont pas été proposé par son département.

```
SELECT student.name , sum(credits) from student , course , takes where
takes.course_id=course.course_id and student.id=takes.id and student.dept_name
!=course.dept_name group by student.name ;
```

```
1 SELECT student.name , sum(credits) from student , course , takes where takes.course_id=course.course_id and student.id=takes.id and student.dept_name !=course.dept_name group by student.name ;
```

NAME	SUM(CREDITS)
Levy	11
Bourikas	7

2 rows returned in 0.01 seconds [Download](#)

24. Pour chaque département, on affiche le nombre total de crédits des cours qui ont eu lieu dans ce département.

```
1 SELECT section . building , sum ( course . credits ) FROM section , course
2 WHERE section . course_id = course . course_id GROUP BY section . building ;
```

Results

ExplainDescribeSaved SQLHistory

BUILDING	SUM(COURSE.CREDITS)
Watson	10
Packard	14
Taylor	17
Painter	11

4 rows returned in 0.04 secondsDownload